WORKED SOLUTION

STEP I 2013, Question 13

Probability and Statistics • Counting and Expectation

STEP I 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q13

The key idea

The numbers $1, \ldots, 52$ split into 26 disjoint "partner pairs" $\{k, 53-k\}$, and a discarded pair is just a partner pair that we have chosen both members of. So to count selections with exactly k complete pairs we choose which pairs are complete, then fill the remaining places with numbers that come from *different* pairs, with a free choice of which partner each time. For the expectation, linearity over the 26 pairs gives a quick confirmation.

Solution

Counting the selections

Write the integers as 26 pairs $\{1,52\},\{2,51\},\ldots,\{26,27\}$. Two integers from 1 to 52 sum to 53 exactly when they form one of these pairs, and each integer lies in exactly one pair. So X is the number of pairs both of whose members are among the seven chosen. Since $2X \leq 7$, X takes the values $0, 1, 2, 3$.

All $\binom{52}{7}$ selections are equally likely, so we count the selections with $X = k$.

- Choose the k complete pairs: $\binom{26}{k}$ ways. These use $2k$ of the seven numbers.
- The other $7-2k$ numbers must come from $7-2k$ *different* pairs among the remaining $26-k$ (two from the same pair would make another complete pair): $\binom{26-k}{7-2k}$ ways.
- From each of those pairs we take one of its two members: 2^{7-2k} ways.

Each selection with $X = k$ arises exactly once this way, so

$$\mathrm{P}(X=k) = \frac{\binom{26}{k}\binom{26-k}{7-2k}\,2^{7-2k}}{\binom{52}{7}}, \qquad k = 0,1,2,3.$$

The numbers

$$\begin{aligned} X=0: &\quad \tbinom{26}{7}2^7 = 657\,800\times 128 = 84\,198\,400,\\ X=1: &\quad 26\tbinom{25}{5}2^5 = 26\times 53\,130\times 32 = 44\,204\,160,\\ X=2: &\quad \tbinom{26}{2}\tbinom{24}{3}2^3 = 325\times 2024\times 8 = 5\,262\,400,\\ X=3: &\quad \tbinom{26}{3}\tbinom{23}{1}2 = 2600\times 23\times 2 = 119\,600, \end{aligned}$$

and $\binom{52}{7} = 133\,784\,560$. These four counts do add up to $\binom{52}{7}$, a good check that nothing has been missed or double counted.

? **Ask yourself** How do we simplify fractions with nine-digit numbers in them?

The note in the question is the hint: $5593 = 7\times 17\times 47$ should be the denominator. Indeed $\binom{52}{7} = 23\,920\times 5593$, and every one of the four counts is a multiple of 23 920:

$$84\,198\,400 = 3520\times 23\,920, \qquad 44\,204\,160 = 1848\times 23\,920,$$
$$5\,262\,400 = 220\times 23\,920, \qquad 119\,600 = 5\times 23\,920.$$

So the distribution of X is

k	0	1	2	3
$\mathrm{P}(X=k)$	$\frac{3520}{5593}$	$\frac{1848}{5593}$	$\frac{220}{5593}$	$\frac{5}{5593}$

(the second is $\frac{264}{799}$ in lowest terms), and $3520+1848+220+5 = 5593$, as it must.

The expectation

$$\mathrm{E}(X) = \frac{0\times 3520 + 1\times 1848 + 2\times 220 + 3\times 5}{5593} = \frac{2303}{5593}.$$

Now $2303 = 7^2\times 47$ and $5593 = 7\times 17\times 47$, so

$$\mathrm{E}(X) = \frac{7^2\times 47}{7\times 17\times 47} = \frac{7}{17},$$

as required.

A quicker route to the expectation

For $j = 1,\ldots,26$ let $I_j = 1$ if both members of the jth pair are chosen and 0 otherwise, so $X = I_1 + \cdots + I_{26}$. Choosing both members of a given pair leaves 5 more from the other 50, so

$$\mathrm{P}(I_j = 1) = \frac{\binom{50}{5}}{\binom{52}{7}} = \frac{7\times 6}{52\times 51} = \frac{7}{442}.$$

By linearity of expectation (which needs no independence; the I_j are in fact dependent),

$$\mathrm{E}(X) = 26\times\frac{7}{442} = \frac{7}{17}.$$

Checks

- The four probabilities sum to 1, and the two methods give the same $\mathrm{E}(X) = \frac{7}{17} \approx 0.412$.
- A computer simulation of 300 000 random selections gave relative frequencies of about 0.630, 0.329, 0.039, 0.001 for $X = 0, 1, 2, 3$, against the exact values 0.6294, 0.3304, 0.0393, 0.0009.
- Sense check: $\mathrm{P}(X = 3)$ is tiny, as three complete pairs in only seven numbers ought to be.

Common slips

- Counting the selections with $X = 0$ as $\binom{52}{7}$ minus something rough, or as $\binom{26}{7}$ alone, forgetting the factor 2^7 for which member of each pair is taken.
- For $X = 1$, choosing the remaining five numbers as $\binom{50}{5}$: that allows a second complete pair and overcounts.

Beyond STEP

Writing a count as a sum of indicator variables and using linearity of expectation is one of the most useful tricks in first-year probability: it gives expectations even when the distribution is hopeless to find. Here the full distribution was within reach, but for, say, the expected number of fixed points of a random permutation, indicators are the natural way in. Blitzstein and Hwang, *Introduction to Probability*, uses this idea again and again.

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